

Combinatorics Foundations

Shel Scholars Competitive Math Academy

Week 0

Introduction

This handout covers the counting foundations that we will assume throughout the AMC 10 Get 100 Camp.

Combinatorics on the AMC 10 is not mostly a matter of clever tricks. It is mostly a matter of being able to answer, without hesitation, questions of the form *how many*. How many numbers are in this list. How many ways can these two independent choices be made. How many arrangements does this collection of objects have. Almost every counting problem on the contest reduces, at its final step, to one of these.

Week 0 has three parts:

1. **Counting the number of numbers in a list.** Inclusive counting, counting multiples of k in a range, and recognizing a list when the problem never calls it one.
2. **The AND and OR principle.** When to multiply, when to add, and how to tell which situation you are in.
3. **Basics of counting and permutations.** The multiplication principle, factorial notation, and arrangements of distinct objects.

Most of this should not be completely new. You have counted things before. Our goal is to make these counts reliable enough that they never become the reason you miss an AMC 10 problem, and to make the underlying moves explicit enough that you can spot them when a problem hides them.

That second goal matters more than it sounds. Many early AMC 10 counting problems do not require any advanced idea at all. They take an elementary count and present it in an unfamiliar form: a list that starts in a strange place, a set of objects that has to be put in order before it can be counted, or a choice that is described in words rather than as a choice.

Main idea/Fact (What Week 0 is for)

Week 0 is prerequisite review, not the main combinatorics curriculum of the camp.

The goal is not merely to know these ideas. The goal is to be able to use them quickly, accurately, and without them getting in the way of the harder counting that comes later.

Counting the Numbers in a List

The Problem

In the AMC 10 you sometimes meet a problem like the following.

Example 1 (2023 AMC 10A, Problem 12)

How many three-digit positive integers N satisfy both of the following properties?

- The number N is divisible by 7.
- The number formed by reversing the digits of N is divisible by 5.

(A) 13 (B) 14 (C) 15 (D) 16 (E) 17

Solution. A number is divisible by 5 only when its last digit is 0 or 5. The *reversed* number ends in the hundreds digit of N , so that hundreds digit must be 0 or 5. A three-digit number cannot start with 0, so N starts with 5. Therefore

$$N \in \{500, 501, \dots, 599\}, \quad 7 \mid N.$$

The multiples of 7 in this range run from $504 = 7 \cdot 72$ to $595 = 7 \cdot 85$:

$$504, 511, 518, \dots, 595.$$

Now the whole problem has collapsed into a single question: *how many numbers are in this list?* And here is the move that finishes it:

504	511	518	...	595
$\div 7$	$\div 7$	$\div 7$	...	$\div 7$
72	73	74	...	85
-71	-71	-71	...	-71
1	2	3	...	14

So there are 14 such integers, and the answer is

(B) 14.

Remark. The clever part of that problem was the divisibility setup. But notice that the *last* step, turning a messy list into $1, 2, \dots, n$, is the entire point of this lecture. That single skill closes dozens of counting problems, and it is worth being able to do it without thinking.

Introduction

This part answers one deceptively simple question:

Given a list of numbers, how many numbers are in it?

For a short list you can just count. But competition lists are long, they skip, they start in strange places, and sometimes they are hidden inside a word problem. Counting them one by one is slow and unreliable.

The whole strategy rests on one observation.

Definition (The anchor list)

The list

$$1, 2, 3, \dots, n$$

has exactly n numbers. This list *counts itself*, and it is the list we will always try to reach.

Main idea/Fact (Main habit for counting a list)

Do not count the list you were given. Transform it, by applying the same operation to *every* term, until it becomes the anchor list $1, 2, \dots, n$. Then the answer is just n .

The reason this is allowed is the most important principle of the lecture.

Main idea/Fact (**PRINCIPLE**: Relabeling)

If you apply the same *reversible* operation to every number in a list, such as adding a constant, subtracting a constant, multiplying or dividing by a nonzero constant, taking a square root, or taking a cube root, then the number of numbers in the list does not change.

Each such operation renames the items without merging or splitting any of them, so the count is untouched. The rest of this part is learning to *spot which operations* carry a given list back to $1, 2, \dots, n$.

Lists that start in the wrong place

Example 2

How many numbers are in the list

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14?$$

Solution. This *is* the anchor list, with $n = 14$, so it has 14 numbers. Nothing to do.

Example 3

How many numbers are in the list

$$9, 10, 11, \dots, 33?$$

Solution. The step is 1, so the only problem is that the list does not start at 1. We slide the whole list down by subtracting 8 from every term:

$$\begin{array}{cccccc} 9 & 10 & 11 & \dots & 33 \\ -8 & -8 & -8 & \dots & -8 \\ \hline 1 & 2 & 3 & \dots & 25 \end{array}$$

This is the anchor list up to 25, so the original list also has

$$\boxed{25}$$

numbers.

Pitfall

A tempting shortcut is to say that $9, 10, \dots, 33$ has $33 - 9 = 24$ numbers. It does not. Subtracting the endpoints counts the *gaps* between the numbers, not the numbers themselves.

That observation deserves a name, because it is the single most common source of off-by-one errors in all of counting.

Main idea/Fact (Inclusive counting)

The list of consecutive integers from a to b , including both endpoints, contains

$$b - a + 1$$

numbers.

The $+1$ counts the starting number itself. Leaving it out is called a **fence-post error**, because n fence posts in a row have only $n - 1$ gaps between them.

Check it against the example: $33 - 9 + 1 = 25$, which agrees. Use the relabeling method to *get* the answer and this formula to *check* it. Two independent routes to the same number is how you catch an off-by-one before it costs you a problem.

Exercise 4. How many numbers are in the list $7, 8, 9, \dots, 41$?¹

Lists that skip

Now the step is bigger than 1.

Example 5

How many numbers are in the list

$$3, 6, 9, \dots, 48?$$

Solution. These are the multiples of 3. To turn a step of 3 into a step of 1, **divide every term by 3**:

$$\begin{array}{cccccc} 3 & 6 & 9 & \dots & 48 \\ \div 3 & \div 3 & \div 3 & \dots & \div 3 \\ \hline 1 & 2 & 3 & \dots & 16 \end{array}$$

That is the anchor list up to 16, so there are

$$\boxed{16}$$

numbers.

Example 6

How many numbers are in the list

$$12, 16, 20, \dots, 60?$$

¹Subtracting 6 from every term gives $1, 2, \dots, 35$, so there are 35. Check: $41 - 7 + 1 = 35$.

Solution. The step is 4, so divide every term by 4:

$$\begin{array}{rcccccc} 12 & 16 & 20 & \cdots & 60 \\ \div 4 & \div 4 & \div 4 & \cdots & \div 4 \\ \hline 3 & 4 & 5 & \cdots & 15 \end{array}$$

Now the step is 1 but the list starts at 3, not 1. So we apply a *second* operation and subtract 2 from every term:

$$\begin{array}{rcccccc} 3 & 4 & 5 & \cdots & 15 \\ -2 & -2 & -2 & \cdots & -2 \\ \hline 1 & 2 & 3 & \cdots & 13 \end{array}$$

So there are

13

numbers.

Main idea/Fact (PRINCIPLE: Divide to collapse the step, then shift to anchor)

For a list with a constant step d :

1. Divide every term by d to make the step 1.
2. Add or subtract a constant so that the list starts at 1.

You may chain as many operations as you need.

This is exactly how a real AMC arithmetic-sequence problem works.

Example 7 (2015 AMC 10A, Problem 7)

How many terms are there in the arithmetic sequence

$$13, 16, 19, \dots, 70, 73?$$

(A) 20 (B) 21 (C) 24 (D) 60 (E) 61

Solution. The step is 3. We subtract 10 from each term, which sets up a clean multiple of 3, and then divide by 3:

$$\begin{array}{rcccccc} 13 & 16 & 19 & \cdots & 73 \\ -10 & -10 & -10 & \cdots & -10 \\ \hline 3 & 6 & 9 & \cdots & 63 \\ \div 3 & \div 3 & \div 3 & \cdots & \div 3 \\ \hline 1 & 2 & 3 & \cdots & 21 \end{array}$$

So the sequence has

(B) 21

terms.

Word problems become list problems

Often the list is described rather than handed to you. The first job is to *build the list*.

Example 8

How many integers between 61 and 305 are divisible by 7?

Solution.

Step 1: find the first and last terms. Divide the endpoints by 7:

$$61 \div 7 = 8 \text{ remainder } 5, \quad 305 \div 7 = 43 \text{ remainder } 4.$$

So the first multiple of 7 at or above 61 is $7 \cdot 9 = 63$, and the last at or below 305 is $7 \cdot 43 = 301$.

Step 2: write the list and count it. The list is

$$63, 70, 77, \dots, 301,$$

which is now an ordinary skipping list. Divide by 7, then subtract 8:

63	70	77	...	301
$\div 7$	$\div 7$	$\div 7$	...	$\div 7$
9	10	11	...	43
-8	-8	-8	...	-8
1	2	3	...	35

So there are

$$\boxed{35}$$

such integers.

Main idea/Fact (The fence-post count)

A list with first term a , last term ℓ , and constant step d , where d divides $\ell - a$, contains

$$\frac{\ell - a}{d} + 1$$

numbers.

The fraction counts the gaps and the +1 counts the starting term. Setting $d = 1$ recovers inclusive counting, $\ell - a + 1$.

Check the example: $\frac{301-63}{7} + 1 = 34 + 1 = 35$.

Main idea/Fact (Counting multiples of k in a range)

The number of multiples of k from 1 to n is

$$\left\lfloor \frac{n}{k} \right\rfloor.$$

The number of multiples of k from a to b inclusive is

$$\left\lfloor \frac{b}{k} \right\rfloor - \left\lfloor \frac{a-1}{k} \right\rfloor.$$

Use this as a second opinion rather than as your primary method. The list method survives contact with problems where the terms are not multiples at all, and the formula does not.

Pitfall

The word “between” is ambiguous about its endpoints. Before counting, check whether the endpoints themselves belong to the list.

In the example above it made no difference, because neither 61 nor 305 is a multiple of 7. When the endpoint *is* a term, decide deliberately whether to include it, and expect the wrong choice to appear among the answer choices.

Exercise 9. How many multiples of 6 are there from 1 to 200?²

Lists that dip below zero

Nothing changes in principle. To escape the negatives you *add* instead of subtracting.

Example 10

How many numbers are in the list

$$-5, -4, -3, \dots, 56, 57, 58?$$

Solution. The step is 1 and the list starts at -5 . To start at 1 we must add 6 to every term:

$$\begin{array}{cccccc} -5 & -4 & -3 & \cdots & 58 \\ +6 & +6 & +6 & \cdots & +6 \\ \hline 1 & 2 & 3 & \cdots & 64 \end{array}$$

So there are

$$\boxed{64}$$

numbers.

Example 11

How many numbers are in the list

$$-9, -6, -3, \dots, 54, 57, 60?$$

Solution. The step is 3 and the list starts at -9 . Add 12 to clear the negatives, landing on a clean multiple of 3, and then divide by 3:

$$\begin{array}{cccccc} -9 & -6 & -3 & \cdots & 60 \\ +12 & +12 & +12 & \cdots & +12 \\ \hline 3 & 6 & 9 & \cdots & 72 \\ \div 3 & \div 3 & \div 3 & \cdots & \div 3 \\ \hline 1 & 2 & 3 & \cdots & 24 \end{array}$$

So there are

$$\boxed{24}$$

numbers.

²The list is 6, 12, $\dots$, 198, and dividing by 6 gives 1, 2, $\dots$, 33, so there are 33. Check: $\lfloor 200/6 \rfloor = 33$.

Pitfall

Do not add just enough to reach 0. The anchor list starts at 1, and 0 is not in it.

Adding 9 to the list above would give $0, 3, 6, \dots$, and dividing by 3 would give $0, 1, 2, \dots, 23$, which has 24 terms rather than 23. Counting that list as 23 is the fence-post error wearing a disguise.

Exercise 12. How many numbers are in the list $-30, -24, -18, \dots, 36, 42$?³

Lists in disguise

The deepest version of the skill is noticing a list when the problem never calls it one.

Example 13

How many perfect squares are there between 5 and 555?

(A *perfect square* is an integer of the form k^2 . For instance $9 = 3^2$ is a perfect square and 8 is not.)

Solution.

Build the list. The smallest perfect square above 5 is $9 = 3^2$. For the top end we estimate:

$$20^2 = 400 < 555, \quad 30^2 = 900 > 555,$$

so the answer is between 20 and 30. Narrowing:

$$24^2 = 576 > 555, \quad 23^2 = 529 < 555.$$

So the list is

$$3^2, 4^2, 5^2, \dots, 23^2.$$

Count it. Here the right reversible operation is the **square root**. Apply it to every term, then subtract to anchor:

3^2	4^2	5^2	$\dots$	23^2
$\sqrt{}$	$\sqrt{}$	$\sqrt{}$	$\dots$	$\sqrt{}$
$\hline 3$	$\hline 4$	$\hline 5$	$\hline \dots$	$\hline 23$
-2	-2	-2	$\dots$	-2
$\hline 1$	$\hline 2$	$\hline 3$	$\hline \dots$	$\hline 21$

So there are

$$\boxed{21}$$

perfect squares between 5 and 555.

Pitfall

When you hunt for the largest term by estimating, always check the *next* candidate too. Here $23^2 = 529 < 555$ but $24^2 = 576 > 555$, so 23^2 really is the last one.

Stopping one term early, or going one term too far, is the most common error in this kind of problem.

³Adding 36 gives $6, 12, \dots, 78$, and dividing by 6 gives $1, 2, \dots, 13$, so there are 13. Check: $\frac{42 - (-30)}{6} + 1 = 13$.

Example 14

How many triples of three consecutive positive integers have a product less than 1,000,000?

Solution. The smallest triple is 1, 2, 3, with product 6. For the largest, write the triple as $n - 1, n, n + 1$, so the product is

$$(n - 1)n(n + 1) = n^3 - n,$$

which is roughly n^3 . Since $100^3 = 1,000,000$, we test near $n = 100$:

$$99 \cdot 100 \cdot 101 = 999,900 < 1,000,000,$$

$$100 \cdot 101 \cdot 102 = 1,030,200 > 1,000,000.$$

So the largest valid triple starts at 99. Now *index each triple by its first integer*, which is what dividing each product by its other two factors accomplishes:

$$\begin{array}{ccccccc} 1 \cdot 2 \cdot 3 & 2 \cdot 3 \cdot 4 & \cdots & 99 \cdot 100 \cdot 101 \\ \div (2 \cdot 3) & \div (3 \cdot 4) & \cdots & \div (100 \cdot 101) \\ \hline 1 & 2 & \cdots & 99 \end{array}$$

So there are

$$\boxed{99}$$

such triples.

Main idea/Fact (PRINCIPLE: Anything you can line up in order is a list)

If the objects you are counting can be put in increasing order and tagged $1, 2, 3, \dots$, then they form a list.

Find the operation, whether a root, a division, or an indexing, that strips each object down to its tag. Then count the tags.

Exercise 15. How many perfect squares are there from 200 to 900 inclusive?⁴

⁴Since $14^2 = 196$ and $15^2 = 225$, the smallest is 15^2 ; since $30^2 = 900$ and $31^2 = 961$, the largest is 30^2 . Square roots give 15, 16, $\dots$, 30, which has $30 - 15 + 1 = 16$ terms.

Summary

Main idea/Fact (Counting checklist)

For counting the number of numbers in a list:

- If the list is hidden in words, **build the list first** by finding its smallest and largest terms.
- If the step is $d > 1$, **divide every term by d** .
- If the list starts in the wrong place, **add or subtract a constant** to anchor it at 1.
- If the terms are squares or cubes, **take the square root or cube root** of every term.
- If the terms are compound objects, **index them** by whichever piece increases by 1 each time.
- You may **chain operations** until you reach $1, 2, \dots, n$. Then the answer is n .

Main idea/Fact (Confirm these)

It might be easy to make mistakes with these things:

- Check the endpoints. The largest term must be in range and the *next* one must be out of range.
- Confirm with the fence-post count $\frac{\ell - a}{d} + 1$, and never forget the $+1$.
- Your relabeled list must start at 1, not at 0. A list running $0, 1, \dots, m$ has $m + 1$ terms.
- If a division does not come out to a whole number, you have misidentified either the step or an endpoint.
- Reread the question. It may ask for something the list only helps you find, such as the largest term or a digit of the count.

Problems

The first several are drills. The last three are real AMC problems. Try as many as you can before looking at the hints.

1. How many numbers are in the list

$$23, 24, 25, \dots, 199, 200?$$

([Hint 1](#), [Hint 2](#))

2. How many numbers are in the list

$$8, 12, 16, \dots, 116, 120?$$

([Hint 1](#), [Hint 2](#))

3. How many integers between 50 and 500 are divisible by 9?

([Hint 1](#), [Hint 2](#))

4. How many numbers are in the list

$$-12, -8, -4, \dots, 96, 100?$$

([Hint 1](#), [Hint 2](#))

5. How many numbers are in the list

$$-20, -15, -10, \dots, 80, 85?$$

([Hint 1](#), [Hint 2](#))

6. How many perfect squares lie strictly between 100 and 1000?

([Hint 1](#), [Hint 2](#))

7. How many perfect cubes lie strictly between 50 and 50000?

([Hint 1](#), [Hint 2](#))

8. How many numbers are in the list

$$3, 7, 11, 15, \dots, 2019, 2023?$$

([Hint 1](#), [Hint 2](#))

Main idea/Fact (Useful for the next problems)

If a prime p divides a perfect square n^2 , then p divides n , and therefore p^2 divides n^2 .

9. (2023 AMC 10A #3) How many positive perfect squares less than 2023 are divisible by 5?

(A) 8 (B) 9 (C) 10 (D) 11 (E) 12

([Hint 1](#), [Hint 2](#))

10. (2024 AMC 10B #3) For how many integer values of x is $|2x| \leq 7\pi$?

(A) 16 (B) 17 (C) 19 (D) 20 (E) 21

([Hint 1](#), [Hint 2](#))

11. (2020 AMC 10B #7) How many positive even multiples of 3 less than 2020 are perfect squares?

(A) 7 (B) 8 (C) 9 (D) 10 (E) 12

([Hint 1](#), [Hint 2](#))

Hints

Hint 1

1. **MAIN IDEA:** This list has step 1, so it just starts in the wrong place. Subtract the same number from every term to slide it down until it begins at 1.

What number must you subtract?

2. **MAIN IDEA:** The step is 4, not 1, so a single shift will not anchor it. Divide every term by 4 first to collapse the step, and only then add or subtract a constant.

What list do you get after dividing?

3. **MAIN IDEA:** Nothing can be counted until the list exists, so build it first. Find the smallest and the largest multiple of 9 in range, and then you are back to an ordinary step-9 list.

What are the first and last terms?

4. **MAIN IDEA:** The list starts negative, so *add* rather than subtract. Choose what you add so that the first term lands on a clean multiple of the step 4, and then divide.

What should the first term become?

5. **MAIN IDEA:** Step 5 and a negative start. Add a constant that clears the negatives and lands the first term on a multiple of 5, then divide every term by 5.

Remember that the anchor list starts at 1, not at 0. What must the first term become?

6. **MAIN IDEA:** This is a list in disguise. Write out the terms as squares, and then take the *square root* of every term to relabel it.

The word “strictly” matters here. Is 100 itself in the list?

7. **MAIN IDEA:** Same disguise, one power higher. Write the list as cubes and take the *cube root* of every term.

The estimation is harder than in the previous problem, so pin down the top end carefully. What is 37^3 ?

8. **MAIN IDEA:** The step is 4. Add the same number to every term to reach a clean multiple of 4, then divide every term by 4 to collapse the step to 1.

What is the smallest number you can add so that 3 becomes a multiple of 4?

9. **MAIN IDEA:** First figure out which squares are even eligible. If 5 divides a perfect square, then 5 has to appear in the prime factorization an even number of times, so the square is divisible by 25.

That means every term of the list is the square of a multiple of 5. What does the list look like?

10. **MAIN IDEA:** An inequality on $|x|$ describes a list of consecutive integers, so find its two endpoints. Divide through by 2 and estimate, using $\pi \approx 3.14159$. You should be thinking about is this way $\pi < 3.1416$ when estimating what the largest integer is.

What is the largest integer x that works?

11. **MAIN IDEA:** The number is a perfect square n^2 that must be divisible by both 2 and 3. As in Problem 9, each prime dividing a square must divide it an even number of times.

What does that force about n itself?

Hint 2

1. Subtracting 22 from every term gives the anchor list

$$1, 2, 3, \dots, 178.$$

Confirm it with inclusive counting, $200 - 23 + 1$.

2. Dividing by 4 gives $2, 3, \dots, 30$, which still starts one place too high, so subtract 1 to reach $1, 2, \dots, 29$.

Confirm it with the fence-post count $\frac{120-8}{4} + 1$.

3. The first multiple is $54 = 9 \cdot 6$ and the last is $495 = 9 \cdot 55$, since $9 \cdot 56 = 504$ is out of range. Dividing the list by 9 gives $6, 7, \dots, 55$, and subtracting 5 anchors it.

How many terms does the anchored list have?

4. Adding 16 to every term gives $4, 8, 12, \dots, 116$, and dividing that by 4 gives the anchor list.

Where does it end?

5. Adding 25 to every term gives $5, 10, 15, \dots, 110$, and dividing by 5 gives the anchor list.

Check your count against $\frac{85 - (-20)}{5} + 1$.

6. Since $10^2 = 100$ is excluded and $32^2 = 1024$ is too large, the list runs from $11^2 = 121$ to $31^2 = 961$. Taking square roots gives $11, 12, \dots, 31$.

How many numbers is that?

7. Since $3^3 = 27$ is too small and $37^3 = 50653$ is too large, the list runs from $4^3 = 64$ to $36^3 = 46656$. Taking cube roots gives $4, 5, \dots, 36$.

How many numbers is that?

8. Adding 1 to every term gives $4, 8, 12, \dots, 2024$, and dividing by 4 gives the anchor list ending at 506.

Check it with $\frac{2023-3}{4} + 1$.

9. Every term is $(5k)^2 = 25k^2$, and the condition $25k^2 < 2023$ becomes $k^2 < 80.9$, so $k \leq 8$. The list is

$$(5 \cdot 1)^2, (5 \cdot 2)^2, \dots, (5 \cdot 8)^2,$$

and stripping each term down to its k leaves the anchor list.

Where does it end?

10. Since $7\pi < 7 \cdot 3.142 = 21.994 < 22$, the condition is $|x| < 11$, so x runs over

$$-10, -9, \dots, -1, 0, 1, \dots, 9, 10.$$

Adding 11 to every term anchors this list. Do not forget that 0 is one of the values.

How many terms are there?

11. Both 2 and 3 must divide n , so n is a multiple of 6 and every term is $(6k)^2 = 36k^2$. The condition $36k^2 < 2020$ gives $k^2 < 56.1$, so $k \leq 7$, and the list is

$$(6 \cdot 1)^2, (6 \cdot 2)^2, \dots$$

Stripping each term down to its k leaves the anchor list. Where does it end?

The AND and OR Principle

The Problem

In the AMC 10 you are often given counting problems like the following.

Example 16 (2012 AMC 10B, Problem 11)

A dessert chef prepares the dessert for every day of a week starting with Sunday. The dessert each day is either cake, pie, ice cream, or pudding. The same dessert may not be served two days in a row. There must be cake on Friday because of a birthday. How many different dessert menus for the week are possible?

(A) 729 (B) 972 (C) 1024 (D) 2187 (E) 2304

Solution. There are seven decisions to make, one per day, so this looks like a job for multiplication. Let us try the obvious order, left to right, and watch it break.

Sunday is free, so it has 4 options. Monday must differ from Sunday, so it has 3. Tuesday must differ from Monday, so 3 again. Wednesday, likewise, 3. Now Thursday. It must differ from Wednesday, and it must also differ from Friday, which is cake. How many options is that?

It depends. If Wednesday happened to be cake, then Thursday only has to avoid cake, so it has 3 options. If Wednesday was anything else, Thursday must avoid two different desserts, so it has 2. The count at this step depends on *which* choice was made earlier, and the moment that happens we are not allowed to multiply.

The repair is not to work harder. It is to make the decisions in a different order. Friday is the day that is pinned down, so start there and work *outward* in both directions:

Sun $\longleftarrow$ Mon $\longleftarrow$ Tue $\longleftarrow$ Wed $\longleftarrow$ Thu $\longleftarrow$ Fri $\longrightarrow$ Sat

Now every remaining day has exactly one already-decided neighbour, the one on its Friday side, and so every remaining day has exactly 3 options no matter what came before. Six days, 3 options each:

$$3^6 = \boxed{\text{(A) } 729}.$$

Remark. Nothing about the problem changed. Only the order in which we made the decisions changed, and that turned a broken count into a one-line count. Knowing *when* you are allowed to multiply, and how to rearrange a problem until you are, is the whole content of this part.

Introduction

Counting means determining, systematically, the number of ways something can happen. You could of course write out every possibility and count them by hand. But there is a good chance you will forget some or count others twice, and on a list of several hundred you will run out of time long before you run out of possibilities.

So we never write out the list. We break the count into a few simple pieces and combine them. There are only two ways to combine, and the first job on every counting problem is to work out which one you are looking at.

Main idea/Fact (Main habit for counting)

Before counting anything, decide which kind of problem you have.

- A *sequence of choices*: do this, *and* then this, *and* then this. AND means **multiply**.
- A *split into cases*: it is either this, *or* that, *or* that. OR means **add**.

Almost every counting problem on the AMC 10 is built out of these two moves and nothing else.

The OR principle

When the outcomes break into separate cases that cannot happen at the same time, count each case on its own and add the results.

Example 17

To get from town A to town C there are 3 direct roads, and also 2 scenic routes that pass through town B . In how many ways can you travel from A to C ?

Solution. A trip is either direct *or* through B , and no trip is both. So the two cases can simply be added:

$$3 + 2 = 5.$$

Main idea/Fact (PRINCIPLE: Addition (OR))

If the outcomes split into cases with nothing in common, with a_1 ways in the first case, a_2 in the second, and so on up to a_k in the last, then the total is

$$a_1 + a_2 + \cdots + a_k.$$

A useful rule of thumb: when you see the word *or*, think addition.

The principle has one requirement, and it is the requirement that gets violated.

Pitfall

The cases must not overlap. If a single outcome falls into two of your cases, adding counts it twice.

For example, the integers from 1 to 20 that are multiples of 2 or of 3 are *not* $10 + 6 = 16$ in number. The multiples of 6, namely 6, 12, and 18, were counted in both lists, so the true count is 13.

Fixing an overlap systematically is called inclusion and exclusion, and it comes later in the camp. For now, build your cases so that no overlap can occur in the first place.

Main idea/Fact (Good cases)

A case split is safe when it is

- *disjoint*, meaning no outcome belongs to two cases, and
- *exhaustive*, meaning every outcome belongs to at least one case.

Split on something that has exactly one answer for each outcome: the leading digit, whether a number is even or odd, which route was taken, how many objects of some type appear.

The AND principle

Now the second move. When you build a single outcome by making a sequence of choices, and the number of options at each step does not depend on the earlier choices, you multiply the option counts.

Example 18

A café sells lattes in 3 sizes and 5 flavors. How many different lattes, one size and one flavor, are possible?

Solution. Choosing a latte means picking a size *and* a flavor. There are 3 ways to pick the size, and for each of those, 5 ways to pick the flavor. So

$$3 \times 5 = 15.$$

Main idea/Fact (PRINCIPLE: Multiplication (AND))

If a task is carried out by making choices in order, with a_1 ways for the first choice, a_2 for the second, and so on up to a_k for the last, and if each of those counts does not depend on the earlier outcomes, then the whole task can be done in

$$a_1 \cdot a_2 \cdots a_k$$

ways.

A useful rule of thumb: when you see the word *and*, think multiplication.

What the AND principle actually requires

Read the condition in that box again, because it is subtler than it looks, and it is exactly what failed in the opening problem.

The multiplication principle does *not* require that the available options stay the same from step to step. It requires only that the *number* of options stay the same, whatever happened earlier.

Example 19

How many three-digit numbers have three *different* digits? For example 120 counts, but 121 does not.

Solution. Fill the three slots in order.

- **Hundreds digit.** Any of 1 through 9, since a three-digit number cannot begin with 0. That is 9 options.
- **Tens digit.** Any digit except the hundreds digit. One nonzero digit is now forbidden, but 0 is allowed here, so again 9 options.
- **Units digit.** Any digit except the two already used, so 8 options.

By the multiplication principle,

$$\underbrace{9}_{\text{hundreds}} \times \underbrace{9}_{\text{tens}} \times \underbrace{8}_{\text{units}} = 648.$$

Why is multiplying legitimate here, when the options at the tens slot clearly depend on what was chosen first? Because the *number* of forbidden values is fixed. Exactly one digit is banned at the tens slot and exactly two at the units slot, no matter which specific digits were chosen earlier. The principle never cared which values were available, only how many.

Main idea/Fact (The test for multiplying)

Before multiplying, ask at each step: is this count the same no matter what was chosen before?

If yes, multiply. If no, either reorder the choices until it is, or split into cases first and multiply inside each case.

Pitfall

In the dessert problem, Thursday had 3 options when Wednesday was cake and 2 otherwise. That is a genuine dependence on history, and multiplying $4 \cdot 3 \cdot 3 \cdot 3 \cdot ?$ was doomed.

Reordering fixed it. When reordering will not fix it, casework will.

Fill the most restricted slot first

The multiplication principle handles any sequence of slot choices, but often one slot carries a restriction the others do not. The fix is small: decide that slot first.

Main idea/Fact (Most restricted first)

When some position carries a special restriction, decide that position *first*. Handling the free slots first often paints you into a corner where the remaining count depends on what you have already done.

Example 20

How many three-digit numbers are there?

Solution. Three slots: hundreds, tens, units. The hundreds digit cannot be 0, or the number would not be three digits long. The other two slots are unrestricted. So handle the hundreds

digit first.

$$\underbrace{9}_{\text{hundreds}} \times \underbrace{10}_{\text{tens}} \times \underbrace{10}_{\text{units}} = 900.$$

For *this* problem the order happened not to matter, since filling units first would still give $10 \times 10 \times 9 = 900$. The leading-digit restriction does not interact with the other slots. As soon as restrictions do interact, order matters, so make handling the restricted slot first an unconditional habit.

Example 21 (2020 AMC 10A, Problem 6)

How many 4-digit positive integers (that is, integers between 1000 and 9999, inclusive) having only even digits are divisible by 5?

(A) 80 (B) 100 (C) 125 (D) 200 (E) 500

Solution. Two restrictions stack here: every digit is even, and the number is divisible by 5. The units digit carries both at once, so start there.

A number is divisible by 5 only when it ends in 0 or 5. But 5 is odd, so the units digit must be 0. That is 1 option, and it is forced.

The thousands digit must be even and cannot be 0, leaving 2, 4, 6, 8, so 4 options. The hundreds and tens digits need only be even, so each has 5 options, namely 0, 2, 4, 6, 8.

$$\underbrace{4}_{\text{thousands}} \times \underbrace{5}_{\text{hundreds}} \times \underbrace{5}_{\text{tens}} \times \underbrace{1}_{\text{units}} = \boxed{\text{(B)} 100}.$$

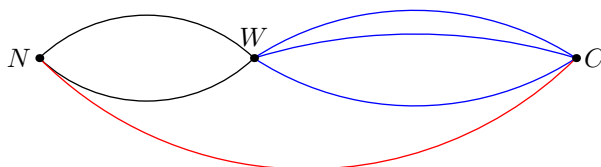
Notice what would have gone wrong starting from the left. You would have said the thousands digit has 4 options, then hesitated at the units digit, because whether it can be 0 has nothing to do with the thousands digit but everything to do with a rule you had not used yet.

Add across cases, multiply within each case

Most real counting problems use both moves at once. You split into cases and add, and inside each case you make a sequence of choices and multiply.

Example 22

You want to travel from New York (N) to Chicago (C). There are 2 roads from N to Washington (W), 3 roads from W to C , and 1 direct road from N to C . In how many ways can you make the trip?



Solution. Split by route type. A trip either goes direct or goes through W , never both, so the cases are disjoint and we may add them.

- **Case 1: direct $N \rightarrow C$.** There is 1 way.

- **Case 2: through W .** This is a two-stage trip, $N \rightarrow W$ and then $W \rightarrow C$, so the multiplication principle gives $2 \times 3 = 6$ ways.

Adding the cases,

$$1 + 6 = 7.$$

Main idea/Fact (The standard shape)

Add the cases, multiply within each case.

Almost every counting problem from here to the end of the camp has this shape.

Example 23

How many four-digit numbers have all four digits even, or all four digits odd?

Solution. No number is both all-even and all-odd, so the two cases are disjoint and we add them. Inside each case we fill four slots and multiply.

All odd. Every digit comes from 1, 3, 5, 7, 9, and none of them is 0, so the leading digit carries no extra restriction:

$$5 \times 5 \times 5 \times 5 = 625.$$

All even. The digits come from 0, 2, 4, 6, 8, but the leading digit cannot be 0, so it has only 4 options:

$$4 \times 5 \times 5 \times 5 = 500.$$

Adding the cases,

$$625 + 500 = 1125.$$

Pitfall

The two cases came out to different numbers, and the reason is the leading zero. Whenever a case involves the digit 0, check the leading slot separately before you multiply.

Summary

Main idea/Fact (Main ideas)

Go through the checklist:

- Decide first: a *sequence of choices* (multiply) or a *split into cases* (add)?
- AND means multiply. OR means add.
- Before multiplying, check that the count at each step is the same no matter what was chosen earlier.
- If the count does depend on history, **reorder** the choices, starting from whatever is already pinned down. If reordering fails, split into cases.
- Fill the most *restricted* slot first, and remember that a number cannot begin with 0.

Main idea/Fact (Double check)

Make sure you don't make silly mistakes by doing the following:

- Check your cases are disjoint. If an outcome could land in two of them, you have counted it twice.
- Check your cases are exhaustive. If an outcome lands in none of them, you have missed it.
- Sanity check the size. A count of arrangements of a handful of objects should not run to the millions, and a count of four-digit numbers can never exceed 9000.
- Reread the question. It may ask for one case rather than the total.

Problems

The first several are drills. The last five are real AMC problems. Try as many as you can before looking at the hints.

1. A café sells a drink by choosing one of 3 sizes, one of 5 flavors, and either hot or iced. How many different drinks are possible?

([Hint 1](#), [Hint 2](#))

2. In how many ways can an international commission be formed if it must contain one European country chosen from among 6, one Asian country from among 4, one North American country from among 3, and one African country from among 7?

([Hint 1](#), [Hint 2](#))

3. How many license plates consist of

(a) 2 letters followed by 3 digits?

(b) 3 letters, followed by 2 even digits, followed by 2 odd digits?

Characters may repeat.

([Hint 1](#), [Hint 2](#))

4. For each of 8 colors you own one shirt and one tie of that color. How many shirt-and-tie outfits can you make if you refuse to wear a shirt and a tie of the same color?

([Hint 1](#), [Hint 2](#))

5. How many subsets does a set of 5 elements have? (Count the empty set and the whole set.)

([Hint 1](#), [Hint 2](#))

6. How many four-digit numbers have all four digits even, or all four digits odd?

([Hint 1](#), [Hint 2](#))

7. (2011 AMC 10A #13) How many even integers are there between 200 and 700 whose digits are all different and come from the set $\{1, 2, 5, 7, 8, 9\}$?

(A) 12 (B) 20 (C) 72 (D) 120 (E) 200

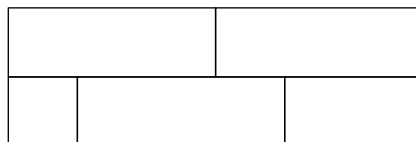
([Hint 1](#), [Hint 2](#))

8. (2022 AMC 10B #3) How many three-digit positive integers have an odd number of even digits?

(A) 150 (B) 250 (C) 350 (D) 450 (E) 550

([Hint 1](#), [Hint 2](#))

9. (2022 AMC 10A #9) A rectangle is partitioned into 5 regions as shown. Each region is to be painted a solid color, red, orange, yellow, blue, or green, so that regions that touch are painted different colors, and colors can be used more than once. How many different colorings are possible?



- (A) 120 (B) 270 (C) 360 (D) 540 (E) 720

([Hint 1](#), [Hint 2](#))

10. (2016 AMC 10A #14) How many ways are there to write 2016 as the sum of twos and threes, ignoring order? (For example, $1008 \cdot 2 + 0 \cdot 3$ and $402 \cdot 2 + 404 \cdot 3$ are two such ways.)

- (A) 236 (B) 336 (C) 337 (D) 403 (E) 672

([Hint 1](#), [Hint 2](#))

11. (2012 AMC 10A #14) Chubby makes nonstandard checkerboards that have 31 squares on each side. The checkerboards have a black square in every corner and alternate red and black squares along every row and column. How many black squares are there on such a checkerboard?

- (A) 480 (B) 481 (C) 482 (D) 483 (E) 484

([Hint 1](#), [Hint 2](#))

Hints

Hint 1

1. **MAIN IDEA:** A drink is built by making three choices one after another, and no choice restricts any other. That is the AND situation.

How many options does each of the three choices carry?

2. **MAIN IDEA:** The commission needs a European country *and* an Asian one *and* a North American one *and* an African one. Every “and” is a multiplication.

Nothing here is a case split, so no addition should appear anywhere in your work. Why not?

3. **MAIN IDEA:** Each character is an independent slot, so count the options at each slot and multiply. In part (b), be careful that “even digit” and “odd digit” are smaller pools than “digit”.

How many even digits are there, counting 0?

4. **MAIN IDEA:** Choose the shirt first, then the tie. Once the shirt is fixed, exactly one tie is forbidden, and that is true no matter which shirt you picked, so multiplication is legal.

How many ties remain after the shirt is chosen?

5. **MAIN IDEA:** Building a subset means making one decision per element: in or out. Five elements means five independent decisions in a row.

How many options does each decision have?

6. **MAIN IDEA:** No number can be both all-even and all-odd, so these are two disjoint cases: count each separately and add. Inside each case you are filling four digit slots, so multiply there.

Which of the two cases has a leading-digit restriction, and why does the other one not?

7. **MAIN IDEA:** Two slots carry restrictions at once. The number must be even, which restricts the units digit, and it must lie between 200 and 700, which restricts the hundreds digit to values available in the given set.

Which digits of $\{1, 2, 5, 7, 8, 9\}$ can be the hundreds digit?

8. **MAIN IDEA:** An odd number of even digits among three means exactly one even digit or exactly three. Rather than grind through both cases, try fixing the hundreds and tens digits first and then asking what the units digit is allowed to do.

Once the first two digits are chosen, is the number of winning units digits always the same?

9. **MAIN IDEA:** This is the most-restricted-slot rule applied to a picture. One region touches more of the others than any region does, so decide that one first, then work outward to its neighbours.

Which region touches the most other regions?

10. **MAIN IDEA:** Translate the words into an equation. With x twos and y threes, $2x + 3y = 2016$ with $x, y \geq 0$, and “ignoring order” means each pair (x, y) is counted once.

Since 2016 and $2x$ are both even, what does that force about $3y$, and therefore about y ?

11. **MAIN IDEA:** Split by row, since the rows come in two kinds. A row that starts with black alternates black, red, black, and a row that starts with red alternates the other way, so the two kinds hold different numbers of black squares.

With 31 corners-to-corner squares and a black square in every corner, which rows start with black?

Hint 2

1. The three counts are 3, 5, and 2, and none of them changes based on the others.

What do you do with a chain of independent choices to get one total?

2. The four counts are 6, 4, 3, and 7. Multiply all four.

There is no “or” anywhere in the problem, which is exactly why nothing gets added.

3. For (a) each letter slot has 26 options and each digit slot has 10, giving $26^2 \cdot 10^3$. For (b) there are 5 even digits, namely 0, 2, 4, 6, 8, and 5 odd ones.

What does that make the product in part (b)?

4. There are 8 shirts, and for each one, 7 acceptable ties.

An alternative route: count all 8×8 outfits and remove the 8 matching ones. Both should agree.

5. Each of the 5 elements is either in or out, so each decision has 2 options, giving 2^5 .

The empty set is the outcome where every decision was “out”, so it is already included.

6. All-odd uses the digits 1, 3, 5, 7, 9 with no leading restriction, giving 5^4 . All-even uses 0, 2, 4, 6, 8, but the leading digit cannot be 0, giving $4 \cdot 5^3$.

Now add the two cases.

7. The hundreds digit must be 2 or 5, and the units digit must be even, so it is 2 or 8. Those two restrictions collide when the hundreds digit is 2, so split into two cases and handle each with multiplication.

In the case where the hundreds digit is 2, how many choices are left for the units digit?

8. Whatever the first two digits are, the parity of the even-digit count is decided entirely by the units digit, and the ten digits split evenly into five even and five odd. So exactly 5 units digits win, every time.

How many ways are there to choose the first two digits, and what do you multiply that by?

9. Start with the wide bottom-middle region, which has 5 colors available. The two top regions must each differ from it and from each other, giving 4 then 3. Each of the two remaining bottom regions touches exactly two already-colored regions, giving 3 each.

Multiply those five counts.

10. Since y must be even, write $y = 2k$, which turns the equation into $2x + 6k = 2016$, so $x = 1008 - 3k$. Each allowed k gives exactly one pair, so you are counting allowed values of k .

The requirement $x \geq 0$ caps k . This is now a list-counting problem of the kind from the first part of this handout.

11. The odd-numbered rows start with black and contain 16 black squares each, while the even-numbered rows start with red and contain 15 each. Among rows 1 through 31 there are 16 odd rows and 15 even ones.

Multiply within each kind of row, then add the two kinds.

Counting and Permutations

The Problem

Here is a problem that needs both principles from the previous part, plus one new idea.

Example 24 (2018 AMC 10A, Problem 4)

How many ways can a student schedule 3 mathematics courses, algebra, geometry, and number theory, in a 6-period day if no two mathematics courses can be taken in consecutive periods? (What courses the student takes during the other 3 periods is of no concern here.)

(A) 3 (B) 6 (C) 12 (D) 18 (E) 24

Solution. Two separate decisions are hiding here, and separating them is the whole problem. First, *which* three periods hold the math classes. Second, *which* class goes in each of those periods.

Which periods. We need three periods from 1, 2, 3, 4, 5, 6, no two consecutive. Listing them systematically by the earliest period used:

$$\{1, 3, 5\}, \quad \{1, 3, 6\}, \quad \{1, 4, 6\}, \quad \{2, 4, 6\}.$$

That is 4 possibilities.

Which class where. The three courses are all different, so once the periods are chosen, the classes can be assigned to them in several orders. Filling the earliest chosen period first, there are 3 choices for it, then 2 for the next, then 1 for the last:

$$3 \times 2 \times 1 = 6.$$

And this count is 6 no matter which set of periods we chose, so the multiplication principle applies across the two stages:

$$4 \times 6 = \boxed{\text{(E) } 24}.$$

Remark. The second stage, counting the orderings of three distinct objects, is the new idea. It comes up so often that it gets its own notation and its own name, and that is what this part is about.

Introduction

An **arrangement**, also called a **permutation**, is an ordering of objects. Counting arrangements uses a product that shrinks by one at each step, which is exactly the multiplication principle from the previous part applied to slots that use up the objects as they go.

Main idea/Fact (Main habit for arrangements)

An arrangement is just a sequence of slot choices, so build it slot by slot and multiply.

The only thing to watch is that each object placed is one fewer object available, so the counts descend: n , then $n - 1$, then $n - 2$, and so on.

Everything in this part assumes that **order matters**, meaning that placing A before B is a different outcome from placing B before A .

Pitfall

Some counting problems deliberately ignore order. A committee, a group, a handshake, or a hand of cards is the same object however you list its members, and those problems are *not* solved by the formulas in this part. They are counted with combinations, which come later in the camp.

Rankings, arrangements, sequences, and anything phrased as first, second, third do keep order, and those are the problems here. Deciding which kind you have is the deepest counting error there is, so ask the question explicitly every time.

Factorials

You met the factorial in the Week 0 Algebra module as a piece of notation. Here is what it counts.

Definition (Factorial)

For a positive integer n , the **factorial** $n!$ is the product

$$n! = n \cdot (n-1) \cdot (n-2) \cdots 3 \cdot 2 \cdot 1,$$

and by convention

$$0! = 1.$$

So $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$. Factorials grow ferociously: $10!$ is already more than three million. That is why you almost never want to compute one in full.

Example 25

Simplify $\frac{10!}{8!}$ and $\frac{42!}{40!}$.

Solution. Expand the larger factorial only as far as the smaller one, then cancel:

$$\frac{10!}{8!} = \frac{10 \cdot 9 \cdot 8!}{8!} = 10 \cdot 9 = 90,$$

$$\frac{42!}{40!} = \frac{42 \cdot 41 \cdot 40!}{40!} = 42 \cdot 41 = 1722.$$

Main idea/Fact (Factorial habit)

When a fraction of factorials appears, expand the top only as far as the bottom and cancel. Never evaluate the factorials separately.

When a *difference* of factorials appears, factor rather than cancel: $8! - 7! = 8 \cdot 7! - 7! = (8 - 1) \cdot 7! = 7 \cdot 7!$.

Arranging all of the objects

Example 26

In how many ways can 4 different books be arranged in a row on a shelf?

Solution. The shelf has four positions, slot 1 through slot 4 from left to right. Arranging the books means deciding which book goes in which slot, and we can do that slot by slot: choose the book for slot 1 *and* then the book for slot 2 *and* so on. Every “and” is the multiplication principle.

- **Slot 1.** Any of the 4 books, so 4 options.
- **Slot 2.** Any book except the one already placed, so 3 options.
- **Slot 3.** Any book except the two already placed, so 2 options.
- **Slot 4.** Only the last remaining book fits, so 1 option.

The *count* at each slot depends only on how many books have been placed, never on which ones, so multiplication is legal:

$$4 \times 3 \times 2 \times 1 = 4! = 24.$$

Main idea/Fact (Arranging n distinct objects)

The number of ways to arrange n distinct objects in a row is

$$n!.$$

Arranging only some of the objects

Example 27

From 5 different books, in how many ways can you choose 3 and arrange them in a row?

Solution. Now there are only 3 ordered slots, but a larger pool of 5 books to draw from.

- **Slot 1.** Any of the 5 books, so 5 options.
- **Slot 2.** Any book except the one already placed, so 4 options.
- **Slot 3.** Any book except the two already placed, so 3 options.

We stop there, because there are only three slots. The remaining two books are simply never shelved.

$$5 \times 4 \times 3 = 60.$$

Notice that this product is a factorial with its tail chopped off:

$$5 \times 4 \times 3 = \frac{5!}{2!} = \frac{5!}{(5-3)!}.$$

Main idea/Fact (PRINCIPLE: Permutations)

The number of ways to choose k of n distinct objects and arrange them in order is

$$\underbrace{n(n-1) \cdots (n-k+1)}_{k \text{ factors}} = \frac{n!}{(n-k)!}.$$

Setting $k = n$ recovers $n!$, since $0! = 1$.

Pitfall

Count the factors, not the endpoints. Arranging 3 of 9 objects gives $9 \cdot 8 \cdot 7$, which is three factors, not $9 \cdot 8 \cdot 7 \cdot 6$.

In practice the descending product is safer than the formula. Write the factors out and stop when you run out of slots.

Arrangements with a restricted position

Nothing new is needed for these. It is the most-restricted-slot rule from the previous part, applied to an arrangement.

Example 28

You have 6 different books, 2 of which are math books. In how many ways can you stack them on a shelf if a math book must be on both ends?

Solution. The two ends carry the restriction, so fill them first.

- **Left end.** Must be a math book, so 2 options.
- **Right end.** Must be the other math book, so 1 option.
- **The middle four slots.** The remaining 4 books in any order, so $4! = 24$ options.

$$2 \times 1 \times 24 = 48.$$

Had we filled the shelf left to right instead, the left end would have had 6 options and we would have been stuck at the right end, where the count would depend on whether a math book had already been used up.

Objects that must stay together

One more standard restriction, and one standard trick for it.

Main idea/Fact (Glue them together)

If two objects must be adjacent, treat them as a single glued block. Then

1. arrange the block together with the remaining objects, and
2. arrange the two objects *inside* the block.

Multiply the two counts, since the second is the same regardless of the first.

Example 29

In how many ways can 5 different books be arranged in a row if two particular books must stand next to each other?

Solution. Glue the two special books into one block. We are now arranging 4 objects, the block and the other three books, which can be done in $4! = 24$ ways. Inside the block the two books can appear in either order, which is $2! = 2$ ways.

$$24 \times 2 = 48.$$

Pitfall

Do not forget the second factor. The block has an inside, and the two objects in it can be swapped.

Problems that ask for the opposite, that two objects must *not* be adjacent, are handled either by listing the acceptable positions directly or by subtracting the adjacent arrangements from the total. The systematic version of that subtraction comes later in the camp.

Summary

Main idea/Fact (Setting up) • Ask first whether *order matters*. If it does not, these formulas do not apply.

- Build an arrangement slot by slot and multiply, with counts descending n , $n - 1$, $n - 2$, and so on.
- Arranging all n distinct objects gives $n!$; arranging k of them gives $n(n - 1) \cdots (n - k + 1) = \frac{n!}{(n - k)!}$.
- Fill restricted positions, such as the ends, *first*.
- If two objects must be adjacent, glue them into a block and remember to arrange inside the block.

Main idea/Fact (Before you bubble) • Count the factors in a descending product against the number of slots. They must match.

- Expand factorial fractions only as far as needed, and factor factorial differences rather than expanding them.
- An arrangement count of n objects can never exceed $n!$, which is a fast upper bound to check against.
- Reread the question. It may ask which objects were chosen rather than how they were ordered.

Problems

The first several are drills. The last four are real contest problems. Try as many as you can before looking at the hints.

1. Compute each of the following without a calculator.

(a) $\frac{9!}{8!}$

(b) $\frac{42!}{40!}$

(c) $\frac{100!}{98!}$

(d) $8! - 7!$

([Hint 1](#), [Hint 2](#))

2. In how many ways can 5 different books be stacked on a shelf?

([Hint 1](#), [Hint 2](#))

3. There are 8 sprinters in the Olympic 100-meter final. The gold medal goes to first place, silver to second, and bronze to third. In how many ways can the medals be awarded?

([Hint 1](#), [Hint 2](#))

4. A club has 15 members. In how many ways can it choose a president, a vice-president, a secretary, and a treasurer, if no member may hold more than one office?

([Hint 1](#), [Hint 2](#))

5. How many three-letter arrangements can be formed from the 26 letters of the alphabet

(a) if letters may repeat?

(b) if no letter may repeat?

([Hint 1](#), [Hint 2](#))

6. You have 6 different books, 2 of which are math books. In how many ways can you stack them on a shelf if you want a math book on both ends?

([Hint 1](#), [Hint 2](#))

7. (2020 AMC 8 #10) Zara has a collection of 4 marbles: an Aggie, a Bumblebee, a Steelie, and a Tiger. She wants to display them in a row on a shelf, but does not want to put the Steelie and the Tiger next to one another. In how many ways can she do this?

(A) 6 (B) 8 (C) 12 (D) 18 (E) 24

([Hint 1](#), [Hint 2](#))

8. (2014 AMC 10A #4) Walking down Jane Street, Ralph passed four houses in a row, each painted a different color. He passed the orange house before the red house, and he passed the blue house before the yellow house. The blue house was not next to the yellow house. How many orderings of the colored houses are possible?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

[\(Hint 1, Hint 2\)](#)

9. (2015 AMC 10A #10) How many rearrangements of $abcd$ are there in which no two adjacent letters are also adjacent letters in the alphabet? For example, no such rearrangements could include either ab or ba .

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

[\(Hint 1, Hint 2\)](#)

10. (2006 AMC 10A #18) A license plate in a certain state consists of 4 digits, not necessarily distinct, and 2 letters, also not necessarily distinct. These six characters may appear in any order, except that the two letters must appear next to each other. How many distinct license plates are possible?

(A) $10^4 \times 26^2$ (B) $10^3 \times 26^3$ (C) $5 \times 10^4 \times 26^2$ (D) $10^2 \times 26^4$ (E) $5 \times 10^3 \times 26^3$

[\(Hint 1, Hint 2\)](#)

Hints

Hint 1

1. **MAIN IDEA:** Never evaluate the factorials separately. For a fraction, expand the top only as far down as the bottom and cancel. For the difference in part (d), factor instead of expanding.

What is the common factor of $8!$ and $7!$?

2. **MAIN IDEA:** Every book gets shelved, so this is an arrangement of all five distinct objects. Build it slot by slot and watch the count descend by one each time.

How many books are available for the last slot?

3. **MAIN IDEA:** Order matters here, since gold, silver, and bronze are different prizes, and only three of the eight runners are placed.

How many slots are you filling, and therefore how many factors should your product have?

4. **MAIN IDEA:** Four distinct offices from a pool of fifteen people, with no repeats. This is the same shape as the sprinters, one size larger.

How many people are still eligible when you fill the fourth office?

5. **MAIN IDEA:** The two parts differ in exactly one respect: whether a letter used in one slot is still available for the next. That is the difference between a constant count and a descending one.

Which of the two parts gives the larger answer, and why should you expect that before computing?

6. **MAIN IDEA:** The two ends carry the restriction, so fill them first and leave the free middle slots for last.

Once a math book is placed on the left end, how many options does the right end have?

7. **MAIN IDEA:** Decide *where* the Steelie and the Tiger go before deciding anything else, since they are the restricted pair. Label the four shelf positions 1, 2, 3, 4 and find which pairs of positions are not adjacent.

How many such pairs of positions are there?

8. **MAIN IDEA:** Start by ignoring the “not next to” rule. Of all the orderings of four houses, each of the two before-rules on its own rules out exactly half.

How many orderings of four distinct houses are there in total?

9. **MAIN IDEA:** There are only $4! = 24$ rearrangements, so the reliable route is an organized search rather than a formula. The forbidden neighbour pairs are ab , bc , and cd , in either order.

Which letter has the fewest forbidden neighbours, and what does that suggest about where to start building?

10. **MAIN IDEA:** The two letters must be adjacent, so glue them into a single block. The plate is then four digits and one block sitting in a row, and there are three independent things to decide.

Treating the block as one unit among five units, how many positions can it occupy?

Hint 2

1. For (a) cancel $8!$ to leave a single factor. For (b) and (c) two factors survive. For (d), write $8! = 8 \cdot 7!$ so that

$$8! - 7! = 8 \cdot 7! - 7! = (8 - 1) \cdot 7!.$$

Now evaluate $7!$ once and multiply.

2. The slot counts are 5, 4, 3, 2, 1, which is exactly $5!$.

What does that evaluate to?

3. Gold has 8 options, silver has 7 once gold is awarded, and bronze has 6. Three slots means three factors.

Multiply $8 \cdot 7 \cdot 6$.

4. The counts descend 15, 14, 13, 12, one per office. In factorial notation this is $\frac{15!}{11!}$.
Multiply the four factors directly rather than computing 15!.

5. With repeats allowed every slot keeps all 26 options, giving 26^3 . With no repeats the counts descend to $26 \cdot 25 \cdot 24$.

The second answer must come out smaller. Check that yours does.

6. The left end has 2 options and the right end then has 1, since only the other math book remains. The four middle slots hold the remaining four books in any order, which is $4!$.

Multiply the three counts.

7. The non-adjacent position pairs are $\{1, 3\}$, $\{1, 4\}$, and $\{2, 4\}$, so 3 pairs. For each pair the Steelie and Tiger can be placed in 2 orders, and the Aggie and Bumblebee then fill the two remaining positions in $2!$ ways.

Multiply those three counts.

8. Of the 24 orderings, requiring orange before red leaves half, and requiring blue before yellow leaves half of those, so 6 survive both. Those six are few enough to write out.

List them, then cross off the ones in which blue and yellow are side by side.

9. Since a is forbidden only next to b , and d only next to c , try building arrangements where b and c are kept apart from their neighbours first, then slot in a and d .

Test each candidate against all three forbidden pairs before accepting it.

10. Three things get decided: the block occupies one of 5 positions among the five units, the four digits are chosen in 10^4 ways, and the two letters in the block are chosen in 26^2 ways, with their order already accounted for by treating them as an ordered pair.

Multiply the three counts, then match the form against the answer choices.

Shef Scholars

Ensuring no Ramanujan or Mirzakhani is ever missed again

